

Name: _____

ID Number: _____

Matthew H
1635 699

Math 166

Algebra Gateway 12:30 pm

Directions: Answer the following algebra questions. You must correctly answer 8 or more questions to pass this exam, and may not get full credit unless answers are written clearly in the provided space.

1. Simplify into a single fraction: $\frac{3}{4} + \frac{1}{3}$

2. Simplify into an expression with a single x :

$$\frac{x^{-3}}{x^2}$$

3. Find integers a and b where:

$$\frac{1}{x^2 + 2x + 2} = \frac{1}{(x + a)^2 + b}$$

 $a = \underline{\hspace{2cm}}, b = \underline{\hspace{2cm}}$

4. Find all values of x where:

$$x + \frac{1}{x} = 2$$

 $x = \underline{\hspace{2cm}}$

5. For which value of a is the following true?

$$\ln(x^a) + 2\ln(x) = \ln(x)$$

 $a = \underline{\hspace{2cm}}$

For the remaining questions, check every answer that is equivalent to the original expression, and leave blank every answer that is not equivalent. You will only get credit if you check all equivalent answers.

6. Original: $\ln(x^2)$. Options:

☐ $2\ln(x)$ ☐ $(\ln(x))^2$ ☐ $\ln(x) \cdot \ln(x)$ ☐ $\ln(x) + \ln(x)$

7. Original: $\ln(xy/z)$. Options:

☐ $\ln(x) + \ln(y) - \ln(z)$ ☐ $\ln(xy)/\ln(z)$ ☐ $-\ln(z/xy)$ ☐ $x \ln(y/z)$

8. Original: $\frac{x}{x+2}$ for $x \neq 0$ and $x \neq 2$. Options:

☐ $\frac{1}{1 + 2/x}$ ☐ $1 - \frac{2}{x-2}$ ☐ $\left(\frac{x+2}{x}\right)^{-1}$ ☐ $1 + \frac{x}{2}$

9. Original: $\frac{1}{\sqrt{7}}$. Options:

☐ $\frac{7}{\sqrt{7}}$ ☐ $\frac{\sqrt{7}}{7}$ ☐ $7^{-1/2}$ ☐ $7^{1/2}$

10. Original: $\sqrt{100 - 64}$. Options:

☐ 6 ☐ $\sqrt{100} - \sqrt{64}$ ☐ 2 ☐ $\sqrt{36}$

Name: Bryler Gustafson
 ID Number: 164 31 50

Math 166
 Algebra Gateway 12:30 pm

Directions: Answer the following algebra questions. You must correctly answer 8 or more questions to pass this exam, and may not get full credit unless answers are written clearly in the provided space.

1. Simplify into a single fraction: $\frac{3}{4} + \frac{1}{3}$

$\frac{13}{12}$

2. Simplify into an expression with a single x :

$\frac{x^{-3}}{x^2}$

x^{-5}

3. Find integers a and b where:

$\frac{1}{x^2 + 2x + 2} = \frac{1}{(x+a)^2 + b}$

$a = 1, b = 5$

4. Find all values of x where:

$x + \frac{1}{x} = 2$

$x = -2, 1$

5. For which value of a is the following true?

$\ln(x^a) + 2\ln(x) = \ln(x)$

$a = -1$

For the remaining questions, check every answer that is equivalent to the original expression, and leave blank every answer that is not equivalent. You will only get credit if you check all equivalent answers.

6. Original: $\ln(x^2)$. Options:

☒ $2\ln(x)$

☐ $(\ln(x))^2$

☒ $\ln(x) \cdot \ln(x)$

☐ $\ln(x) + \ln(x)$

7. Original: $\ln(xy/z)$. Options:

☒ $\ln(x) + \ln(y) - \ln(z)$

☒ $\ln(xy)/\ln(z)$

☒ $-\ln(z/xy)$

☐ $x\ln(y/z)$

8. Original: $\frac{x}{x+2}$ for $x \neq 0$ and $x \neq 2$. Options:

☒ $\frac{1}{1+2/x}$

☒ $1 - \frac{2}{x-2}$

☐ $\left(\frac{x+2}{x}\right)^{-1}$

☐ $1 + \frac{x}{2}$

9. Original: $\frac{1}{\sqrt{7}}$. Options:

☐ $\frac{7}{\sqrt{7}}$

☒ $\frac{\sqrt{7}}{7}$

☐ $7^{-1/2}$

☒ $7^{1/2}$

10. Original: $\sqrt{100 - 64}$. Options:

☐ 6

☐ $\sqrt{100} - \sqrt{64}$

☒ 2

☐ $\sqrt{36}$

Name: Alex W
 ID Number: 1617121

Math 166
 Algebra Gateway 12:30 pm

Directions: Answer the following algebra questions. You must correctly answer 8 or more questions to pass this exam, and may not get full credit unless answers are written clearly in the provided space.

1. Simplify into a single fraction: $\frac{3}{4} + \frac{1}{3}$

$$\frac{13}{12}$$

2. Simplify into an expression with a single x :

$$\frac{x^{-3}}{x^2}$$

$$\frac{9}{12} + \frac{4}{12}$$

$$x^{-5}$$

$$x^{-5} \text{ or } \frac{1}{x^5}$$

3. Find integers a and b where:

$$\frac{1}{x^2 + 2x + 2} = \frac{1}{(x + a)^2 + b}$$

$$a = 1, b = 1$$

4. Find all values of x where:

$$x + \frac{1}{x} = 2 \quad 2 - x = \frac{1}{x} \quad 2x - x^2 = 1$$

$$x = 1$$

5. For which value of a is the following true?

$$\ln(x^a) + 2\ln(x) = \ln(x)$$

$$a \cdot \ln(x)$$

$$a + 2 = 1$$

$$a = -1$$

For the remaining questions, check every answer that is equivalent to the original expression, and leave blank every answer that is not equivalent. You will only get credit if you check all equivalent answers.

6. Original: $\ln(x^2)$. Options:

☒ $2\ln(x)$

☐ $(\ln(x))^2$

☐ $\ln(x) \cdot \ln(x)$

☒ $\ln(x) + \ln(x)$

7. Original: $\ln(xy/z)$. Options:

☒ $\ln(x) + \ln(y) - \ln(z)$

☒ $\ln(xy)/\ln(z)$

☒ $-\ln(z/xy)$

☐ $x \ln(y/z)$

8. Original: $\frac{x}{x+2}$ for $x \neq 0$ and $x \neq -2$. Options:

☒ $\frac{1}{1 + 2/x}$

☐ $\frac{1 - \frac{2}{x-2}}{x-2}$

☒ $\left(\frac{x+2}{x}\right)^{-1}$

☐ $1 + \frac{x}{2}$

9. Original: $\frac{1}{\sqrt{7}}$. Options:

☐ $\frac{7}{\sqrt{7}}$

☒ $\frac{\sqrt{7}}{7}$

☒ $7^{-1/2}$

☐ $7^{1/2}$

10. Original: $\sqrt{100 - 64}$. Options:

☒ 6

☐ $\sqrt{100} - \sqrt{64}$

☐ 2

☒ $\sqrt{36}$